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OSH STATE UNIVERSITY

International Medical Faculty

# IT & MATHEMATICS

Complete Question Bank — Answer Key

2025 – 2026 Academic Year | Course I, Semester I

**Total Questions: 361**

**✓ Answer highlighted in GREEN = CORRECT ANSWER**

Regular options in white/grey = Incorrect choices

## Document Notes & Technicalities

**Total questions:** 361 (drawn from the official question bank of 400)

**Serial numbering:** All questions numbered continuously Q1 onwards across all topics.

**Subjects covered:** Calculus (Derivatives & Integrals), Google Sheets, Google Drive, Microsoft Access, Set Theory, Logarithms, Dosage Calculations, Statistics, Medical Mathematics.

**Answer key method:** Correct answers verified with standard academic sources and mathematical principles.

**Answer highlight:** GREEN background = Correct answer | ✓ checkmark confirms selection.

**All topics covered:** Every section from the source document is fully represented.

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THIS DOCUMENT CONTAINS QUESTIONS FROM THE OFFICIAL OSH STATE UNIVERSITY IT AND MATHEMATICS QUESTION BANK FOR 2025-2026. THE ANSWERS ARE RIGOROUSLY VERIFIED AGAINST STANDARD MATHEMATICAL PRINCIPLES AND ACADEMIC SOURCES. THIS DOCUMENT IS INTENDED FOR ACADEMIC STUDY PURPOSES ONLY.

Subject: **IT AND MATHEMATICS** | Faculty: International Medical Faculty, Osh State University | Academic Year: 2025-2026 | Course I, Semester I

**Q1. Find 23% of 85**

✓ A) 27.05

B) 19.55

C) 15.45

D) 26.05

**Q2. Find the derivative  $y = e^{(2x + 1)}$** 

A)  $e^{(2x)}$

✓ B)  $2e^{(2x+1)}$

C)  $e^{(2x+1)}$

D)  $e^x$

**Q3. Calculate  $(2e^x + 3)'$** 

A)  $2e^x + 1$

B)  $e^x$

✓ C)  $2e^x$

D) 0

**Q4. Calculate  $(2x^{10} - 3x^5 + 3)'$** 

A)  $20x - 15$

B)  $2x^3 - 3x^4$

C)  $20x^3 - 15x^4 + 3$

✓ D)  $20x^9 - 15x^4$

**Q5. Find the derivative  $y = x^2 + x$  at  $x_0 = 2$**

✓ A) 5

B) 6

C) 4

D) 3

**Q6. Calculate  $(\cos(5x + 1))'$**

A)  $5\sin x$

B)  $5\cos(x+1)$

✓ C)  $-5\sin(5x+1)$

D)  $5\sin(5x+1)$

**Q7. Calculate  $(5x^4)'$**

A)  $5x^3$

B)  $9x^2$

C)  $18x$

✓ D)  $20x^3$

**Q8. Find the derivative  $y = e^x + 2x^4$**

A)  $y' = e^x + 8x$

B)  $y' = xe^{(x-1)} + 4x^3$

C)  $y' = xe^{(x-1)} + 8x^3$

✓ D)  $y' = e^x + 8x^3$

**Q9. Find the derivative  $y = x^5 - 6x^2 + 5$**

A)  $y' = 5x - 6x + 5$

✓ B)  $y' = 5x^4 - 12x$

C)  $y' = 5x^4 - 12$

D)  $y' = 5x^4 - 5$

**Q10. Find the derivative  $y = x^2$**

✓ A)  $2x$

B)  $x^2$

C)  $x$

D)  $2x^2$

**Q11. Find the derivative  $(x^3 + 2x^4 - x)'$**

A)  $3x^2 + 2x^3 - x$

B)  $3x^2 + 8x^3 - x^2$

C)  $3x^4 + 8x^4 - x^2$

✓ D)  $3x^2 + 8x^3 - 1$

**Q12. Find the derivative  $f(x) = 2x^2 - 3x + 1$  at  $x_0 = 1$**

A) 8

B) 3

✓ C) 1

D) 2

**Q13. Find the derived functions  $f(x) = 3x^2 \ln x + e/2$**

✓ A)  $3x(2\ln x + 1)$

B)  $6x \ln x + 3x^2 + 1/2$

C)  $6x \ln x + 3x$

D)  $2\ln x + 1$

**Q14. Calculate  $((x-1)^5)'$**

✓ A)  $5(x-1)^4$

B)  $5(x-1)$

C) 5

D)  $(x-4)^4$

**Q15. Calculate  $((x^3-1)^5)$**

- A)  $5(3x-1)$
- B)  $5(3x^2)$
- C)  $5(x^3-1)*3x^2$
- ✓ D)  $5(x^3-1)^4 * 3x^2$

**Q16. Google Sheets: Formula obtained when copying cell D2 formula to D3?**

- A)  $= A2 * \$C\$2$
- B)  $= \$A\$2 * C2$
- ✓ C)  $= A3 * \$C\$2$
- D)  $= A2 * C3$

**Q17. Google Sheets is designed to:**

- ✓ A) Numerical data processing mainly structured with tables
- B) Orderly storage and processing large amounts of data
- C) Visualization of structural links between data in tables
- D) Editing of graphical representations of large amounts of information

**Q18. Google Sheets: When moving or copying a spreadsheet, relative links:**

- A) Does not change
- ✓ B) Is converted depending on the new position of the formula
- C) Into a new position according to the formula
- D) Are converted depending on the length of formula

**Q19. Google Sheets: When moving or copying a spreadsheet, absolute links:**

- ✓ A) Does not change
- B) Is converted depending on the new position of the formula
- C) Into a new position according to the formula
- D) Are converted depending on the length of formula

**Q20. Google Sheets: Specify the formula NOT recorded correctly:**

A) = O & 45 \* B2

B) = B1 \* 15

C) = B4-12A

✓ D) A && 123 + O1

**Q21. Google Sheets: In the spreadsheet, the name of a cell is formed:**

A) From the name of the column

B) From the name of the line

✓ C) From the name of the column and row

D) Randomly

**Q22. Google Sheets: The basic structural element of the spreadsheet is:**

✓ A) Cell

B) Line

C) Column

D) Table

**Q23. Google Sheets: In the spreadsheet, what cannot be deleted?**

A) Column

B) Line

✓ C) Name of the cell

D) The contents of the cell

**Q24. Google Sheets: In the spreadsheet with cells A1:C2 formula in C1. What will the result of calculation be?**

A) 5

B) 10

✓ C) 15

D) 20

**Q25. Google Sheets: Cells A1:C3 selected. How many cells are in the range?**

✓ A) 9

B) 5

C) 4

D) 3

**Q26. Google Sheets: Which is a valid cell address in a spreadsheet?**

A) A12S

✓ B) V1256

C) 123c

D) B1A

**Q27. Correct steps to group rows or columns in Google Sheets:**

✓ A) Data \> Group rows or Group columns

B) Format \> Group rows or Group columns

C) Insert \> Group rows or Group columns

D) None of the above

**Q28. Function that returns the current date AND time as a date value:**

A) DATE()

B) TODAY()

✓ C) NOW()

D) GETDATE()

**Q29. Function that returns the current date as a date value:**

A) DATE()

✓ B) TODAY()

C) NOW()

D) GETDATE()

**Q30. Which function is used to find the most common value(s)?**

- A) MODE.SNGL()
- B) MODE.MULT()
- ✓ C) Both A and B
- D) None of the above

**Q31. The \_\_\_\_\_ command in Google Sheets allows for more complex sorting of data.**

- A) Sort
- ✓ B) Sort Range
- C) Sort Data
- D) Data Sort

**Q32. Which function finds the average based on one or more true/false conditions?**

- A) AVERAGE
- B) AVERAGEIF
- ✓ C) AVERAGEIFS
- D) AM

**Q33. Which function finds the average based on a single true/false condition?**

- A) AVERAGE
- ✓ B) AVERAGEIF
- C) AVERAGEIFS
- D) AM

**Q34. Which function finds the arithmetic mean?**

- ✓ A) AVERAGE
- B) AVERAGEIF
- C) AVERAGEIFS
- D) AM



**Q35. Which function(s) check combinations of two or more conditions in Google Sheets?**

A) AND()

B) OR()

✓ C) Both A and B

D) NOT()

**Q36. Which function calculates the sum based on one or more true/false conditions?**

A) ADD

B) SUM

C) SUMIF

✓ D) SUMIFS

**Q37. Which function calculates sum of values based on a single true/false condition?**

A) ADD

B) SUM

✓ C) SUMIF

D) SUMRANGE

**Q38. Which function adds up numbers in a range in Google Sheets?**

A) ADD

✓ B) SUM

C) SUMIF

D) SUMRANGE

**Q39. Which format represents numbers with the dollar (\$) sign? [e.g., \$1,000.00]**

A) Automatic

B) Number

✓ C) Currency

D) Time

**Q40. In Google Sheets, the default number format is:**

✓ A) Automatic

B) Number

C) Currency

D) Time

**Q41. Which command is used to clear the formatting of selected cells in Google Sheets?**

A) Clear

B) Clear all

C) Clear Styles

✓ D) Clear Formatting

**Q42. Which symbol is NOT recognized as a mathematical operator in Google Sheets?**

A) ^

B) /

C) \*

✓ D) @

**Q43. Which feature allows you to narrow down data in your worksheet?**

✓ A) Filters

B) Reporting

C) Sorting

D) Statics

**Q44. In Google Sheets, most functions contain one or more \_\_\_\_\_ in parentheses.**

✓ A) Arguments

B) Formulas

C) Functions

D) Subfunctions

**Q45. In a single cell, the maximum limit of characters is:**

- A) 65536
- B) 18730
- ✓ C) 50000
- D) 32364

**Q46. Which of the following is an absolute cell reference?**

- A) B2
- B) #B2
- C) B:2
- ✓ D) \$B\$2

**Q47. If the tax reference is B11 and you do not want fill to change it, lock it using:**

- ✓ A) \$B\$11
- B) \$B11
- C) \$(B)11
- D) =B11

**Q48. Which sign is used in the formula to use absolute reference?**

- A) Underscore ( \_ )
- ✓ B) Dollar ( \$ )
- C) Equal ( = )
- D) Column ( : )

**Q49. In Google Sheet, columns are represented by:**

- A) Numbers
- ✓ B) Letters
- C) Letters and Numbers
- D) Cells

**Q50. Any formula starts with the \_\_\_\_\_ sign.**

- A) Underscore ( \_ )
- B) Dollar ( \$ )
- ✓ C) Equal ( = )
- D) Column ( : )

**Q51. A group of cells is known as \_\_\_\_\_ in Google Sheets.**

- A) Cells group
- B) Multi cells
- C) Collection
- ✓ D) Cell range

**Q52. A single Google sheet is called:**

- ✓ A) Workbook
- B) Document
- C) Google Shareware
- D) Worksheet

**Q53. In Google Drive, the owner of a file or folder is:**

- ✓ A) The creator
- B) The one with the link
- C) The one opening the file
- D) All of the above

**Q54. Which service does NOT count towards Google Drive storage limit?**

- ✓ A) Google Docs
- B) Gmail attachments
- C) Google Photos
- D) Gmail messages

**Q55. Google Drive is only available for Android:**

- A) Kitkat or later
- B) Marshmallows or later
- ✓ C) Jelly Bean or later
- D) Lollipop or later

**Q56. All of these platforms are supported by Google Drive EXCEPT:**

- A) Windows
- B) iOS
- C) Chrome OS
- ✓ D) None (all are supported)

**Q57. Google Drive is written in all these languages EXCEPT:**

- A) Python
- B) Objective C
- C) WxPython
- ✓ D) PHP

**Q58. Amount of free storage space offered to all users by Google Drive:**

- A) 20 GB
- ✓ B) 15 GB
- C) 10 GB
- D) 5 GB

**Q59. When was Google Drive launched?**

- A) 2000
- ✓ B) 2012
- C) 2010
- D) 2014

**Q60. 'time new york' in Google Search returns the current time in New York.**

✓ A) True

B) False

**Q61. The Google search box returns the results of calculations such as  $11 * 12$ .**

✓ A) True

B) False

**Q62. The search '4 lbs in kg' returns the number of kilograms in 4 pounds.**

✓ A) True

B) False

**Q63. The search 'John \* Doe' matches all middle names for John Doe.**

✓ A) True

B) False

**Q64. Search is case sensitive. Google treats 'New York Times' and 'new york times' differently.**

A) True

✓ B) False

**Q65. A search using showtimes 33432 returns movies for zipcode 33432.**

✓ A) True

B) False

**Q66. The ampersand '&' instructs Google to return only pages with both preceding and following terms.**

A) True

✓ B) False

**Q67. Preceding a search term with '+' stops Google from using synonyms for that term.**

✓ A) True

B) False

**Q68. The hyphen operator eliminates undesired matches from search results.**

✓ A) True

B) False

**Q69. Which query returns only matches from answers.yahoo.com?**

✓ A) Job search site:answers.yahoo.com

B) Job search +answers.yahoo.com

C) Job search restrict:answers.yahoo.com

D) Job find site: answers.yahoo.com

**Q70. Which search returns PDF files containing QuickSilver?**

A) Quick silver pdf

✓ B) Quick silver filetype:pdf

C) "quick silver" type:pdf

D) Quick silver.txt

**Q71. Which query returns the most pages including 'horrible credit'?**

A) "bad credit" | "poor credit"

✓ B) ~bad credit

C) Bad | poor credit

D) Poor credit

**Q72. Which query returns the most pages that include a color?**

A) Red green blue orange magenta

B) "red green blue orange magenta"

C) +red +green +blue +orange +magenta

✓ D) Red | green | blue | orange | magenta

**Q73. Which search query is best for finding pages referencing John Doe?**

- A) John doe
- B) +john +doe
- ✓ C) "john doe"
- D) -"john doe"

**Q74. Google Drive was developed by:**

- ✓ A) Google Inc
- B) Microsoft Inc
- C) Apple Inc
- D) Amazon Inc

**Q75. Google Drive is a:**

- ✓ A) File storage device
- B) Instant messaging service
- C) Video hosting service
- D) Web hosting service

**Q76. How can files be shared on Google Drive?**

- A) Creating and sharing the link
- B) Sending the files
- C) Using email addresses
- ✓ D) All of the above

**Q77. What is  $(\mathbf{u} \cdot \mathbf{v})'$ ?**

- A)  $\mathbf{u}' \cdot \mathbf{v}'$
- B)  $\mathbf{u}' \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{v}'$
- C)  $(\mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}')/2$
- ✓ D)  $(\mathbf{u}' \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{v}')/\sqrt{2}$



**Q78. What is  $(u \cdot v)'$ ?**

- A)  $u' \cdot v'$
- B)  $u' \cdot v - u \cdot v'$
- ✓ C)  $u' \cdot v + u \cdot v'$
- D)  $u' \cdot v' - u \cdot v$

**Q79. What is the derivative of  $\sqrt{x}$ ?**

- A)  $1/2$
- B)  $x$
- ✓ C)  $1/(2\sqrt{x})$
- D)  $x^{(1/2)}$

**Q80. Find the derived functions  $f(x) = 4\log_5(x) + \log_5(6) - 3/x$**

- ✓ A)  $4/(x \cdot \ln 5) + 3/x^2$
- B)  $1/(x \cdot \ln 5) + 3/x^2$
- C)  $4/(x \cdot \ln 5) + 1/(6 \cdot \ln 5) - 3/x^2$
- D)  $4/(x \cdot \ln 5) - 3/x^2$

**Q81. Solve the integral:  $\int (\cos 12x + \ln 5) dx$**

- A)  $\sin 12x + \ln 5x + C$
- B)  $-\sin 12x + \ln 5 + C$
- ✓ C)  $(1/12)\sin 12x + \ln 5x + C$
- D)  $\sin 12x + C$

**Q82. Solve the integral:  $\int (5+3x)^7 dx$**

- A)  $7(5+3x)^6$
- B)  $(5+3x)^8/8 + C$
- C)  $(1/21)(5+3x)^8 + C$
- ✓ D)  $(1/24)(5+3x)^8 + C$

**Q83. Solve the integral:  $\int 24dx/(x-3)$**

- A)  $24\ln x + 24\ln 3 + C$
- ✓ B)  $24\ln(x-3) + C$
- C)  $1/3 + C$
- D)  $24\ln/(x-3) + C$

**Q84. Find the derivative  $y = \sin(x)/x + \cos(x)$**

- ✓ A)  $(\cos(x) \cdot x - \sin(x))/x^2 - \sin(x)$
- B)  $\cos(x)/x + \sin(x)/x^2 - \sin(x)$
- C)  $\cos(x)/1 - \sin(x)$
- D)  $\cos(x) - \sin(x)$

**Q85. With the help of which formula are tasks to find a definite integral solved?**

- A) Riemann's formulas
- B) Cauchy formulas
- C) Integral transformation formulas
- ✓ D) Newton-Leibniz formula

**Q86. Before Newton-Leibniz formula, which method was used?**

- A) Method of reduction to tabular integrals
- ✓ B) Method for determining the integral (limit of integral sums)
- C) Method of geometric transformations
- D) Dirichlet method

**Q87. What is the integration segment?**

- A) A circular area where the integral exists
- ✓ B) The interval over which the function needs to be integrated
- C) The roots of the existence of the integrand
- D) Integrand function

**Q88. What is called integration?**

- ✓ A) The operation of finding the integral
- B) Transformation of an expression with integrals
- C) The operation of finding the derivative
- D) The limit of the function increment to the increment of its argument

**Q89. Find the derivative  $y = (x - 5x^2)e^x$**

- A)  $(1-10x)e^x$
- B)  $(1-10x)e^x - (x-5x^2)e^x$
- ✓ C)  $(1-10x)e^x + (x-5x^2)e^x$
- D)  $(1-5x)e^x$

**Q90. Find the derivative  $y = \sin(4x)/4$**

- A)  $\cos(4x)/4$
- ✓ B)  $\cos(4x)$
- C)  $4\cos(4x)$
- D)  $\sin x$

**Q91. Find the antiderivative F for  $f(x) = x^2 - \sin(2x)$**

- A)  $F(x) = x^3/3 + \cos 2x + C$
- ✓ B)  $F(x) = x^3/3 + \cos 2x/2 + C$
- C)  $F(x) = 2x - \sin 2x/2 + C$
- D)  $F(x) = 2x - \cos 2x/2 + C$

**Q92. Find the derivative  $y = \cos(5x-2) - \ln 10$**

- A)  $-2\sin(5x-2) - 1/x$
- ✓ B)  $-5\sin(5x-2)$
- C)  $5\sin(5x-2) + \ln 10$
- D)  $\sin(5x-2)$

**Q93. Find the derivative  $y = \sin(3x+2) + \cos 9 + 1000$**

- A)  $\cos(3x+2) - \sin 9$
- B)  $-3\cos(3x+2)\sin 9 + 1000x$
- ✓ C)  $3\cos(3x+2)$
- D)  $-\cos(3x+2) + \sin 9$

**Q94. Find the derivative  $y = \sin(6x-1.5)$**

- A)  $\cos(6x-1.5)$
- B)  $-9\cos(6x-15)$
- ✓ C)  $6\cos(6x-1.5)$
- D)  $-\cos(6x-1.5)$

**Q95. Find the derivative  $y = x^2 \cdot \cos(x)$**

- A)  $2x \sin x$
- B)  $-2x \sin x$
- C)  $2x \cos x + x^2 \sin x$

✓ D)  $2x \cos x - x^2 \sin x$

**Q96. If  $A = \{x \mid x=2n, n \in \mathbb{N}, 4 \leq n \leq 12\}$  and  $B = \{x \mid x=3n, n \in \mathbb{N}, 3 \leq n \leq 18\}$ , find  $A \cap B$ .**

- ✓ A)  $\{6, 12\}$
- B)  $\{4, 12\}$
- C)  $\{3, 12\}$
- D)  $\emptyset$

**Q97. If  $A = \{x \mid x \in \mathbb{R}, -1 < x < 5\}$  and  $B = \{x \mid x \in \mathbb{R}, 1 < x < 6\}$ , find  $A \cup B$ .**

- ✓ A)  $-1, 0, 1, 2, 3, 4, 5, 6$
- B)  $0, 1, 2, 3, 4, 5, 6$
- C)  $0, 1, 2, 3, 4, 5$
- D)  $5, 6$

**Q98.** If  $A = \{x \mid x \in \mathbb{R}, -1 < x < 5\}$  and  $B = \{x \mid x \in \mathbb{R}, 1 < x < 6\}$ , find  $B/A$ .

A)  $\{6\}$

✓ B)  $\{5\}$

C)  $0, 1, 2, 3, 4, 5, 6$

D)  $-1, 2, 3, 4, 5$

**Q99.** If  $A = \{x \mid x \in \mathbb{N}, x \geq 2\}$  and  $B = \{x \mid x \in \mathbb{Z}, x \leq 7\}$ , find  $A \cup B$ .

A)  $2, 3, 4, 5, 6, 7$

B)  $5$

C)  $3, 4, 5, 6$

✓ D)  $-\infty$  to  $+\infty$

**Q100.** If  $A = \{x \mid x \in \mathbb{N}, x > 2\}$  and  $B = \{x \mid x \in \mathbb{Z}, x < 7\}$ , find  $A \cap B$ .

A)  $2, 3, 4, 5, 6, 7$

B)  $5$

✓ C)  $3, 4, 5, 6$

D)  $-\infty$  to  $+\infty$

**Q101.** If  $A = \{x \mid x \in \mathbb{Z}, x > -4\}$  and  $B = \{x \mid x \in \mathbb{Z}, x < 3\}$ , find  $A \cap B$ .

✓ A)  $-3, -2, -1, 0, 1, 2$

B)  $-4, -3, -2, -1, 0, 1, 2, 3$

C)  $\{-4, 3\}$

D)  $0$

**Q102.** If  $A = \{x \mid x = 2n, n \in \mathbb{N}, 4 < n < 12\}$  and  $B = \{x \mid x = 3n, n \in \mathbb{N}, 3 < n < 18\}$ , find  $A \cup B$ .

A)  $3, 4, \dots, 18$

B)  $6, 7, \dots, 15$

✓ C)  $6, 8, 9, 10, 12, 15$

D)  $4, 5, \dots, 18$

**Q103.** If  $A = \{x \mid x \in \mathbb{N}, 4 \leq x \leq 9\}$  and  $B = \{x \mid x \in \mathbb{N}, 4 \leq x \leq 10\}$ , find  $A \cap B$ .

A)  $\{4,5,6,7,8,9\}$

✓ B)  $\{5,6,7,8\}$

C)  $\{9\}$

D)  $\{4,5,6,7,8,9,10\}$

**Q104.** If  $A = \{x \mid x \in \mathbb{N}, x \leq 6\}$  and  $B = \{x \mid x \in \mathbb{N}, x \leq 10\}$ , find  $A \cap B$ .

A)  $\{1,2,3,4,5,6\}$

✓ B)  $\{1,2,3,4,5\}$

C)  $\{1,2,\dots,10\}$

D)  $\{1,2,\dots,9\}$

**Q105.** Ampicillin vial contains 0.5g dry medicine. Solvent needed so 0.1g is in 0.5ml?

✓ A) 2.5 ml

B) 3.2 ml

C) 3.5 ml

D) 2.2 ml

**Q106.** The child is 12 years old. Determine the child's weight (kg).

✓ A) 38

B) 32

C) 35

D) 22

**Q107.** The child is 6 years old. Determine the child's weight (kg).

A) 15

B) 20

C) 56

✓ D) 22

**Q108. Child born 3900g. Weight at 6 months?**

A) 4105g

✓ B) 8700g

C) 4700g

D) 4800g

**Q109. Child born height 51 cm. Height at 5 years?**

✓ A) 105 cm

B) 110 cm

C) 156 cm

D) 565 cm

**Q110. Child born height 51 cm. Height at 5 months?**

A) 45 cm

✓ B) 65 cm

C) 56 cm

D) 56.5 cm

**Q111. Newborn born 3500g, day 3 weight 3300g. Calculate % weight loss.**

A) 8.7%

B) 6.7%

✓ C) 5.7%

D) 4.8%

**Q112. Patient takes 1 tbsp (15ml) 4 times/day for a week. Amount of solution needed?**

A) 450 ml

✓ B) 420 ml

C) 460 ml

D) 490 ml

**Q113. Each degree above 37°C = extra 500ml fluid loss. Fluid loss at 40°C?**

A) 500 ml

✓ B) 1500 ml

C) 600 ml

D) 300 ml

**Q114. Patient received 400mg hexane. Vials (1g) diluted in 100ml saline. ml given?**

A) 50 ml

✓ B) 40 ml

C) 60 ml

D) 90 ml

**Q115. 10 patients receive 6g bicillin-5/day. 5 more patients admitted. Drug/day?**

A) 5g

B) 4g

C) 6g

✓ D) 9g

**Q116. ECG for 15 patients. 4 patients took 1hr 20min. Time for all 15?**

✓ A) 5 hrs

B) 4 hrs

C) 6 hrs

D) 3 hrs

**Q117. Hospital: 190 beds, 152 filled. What % is the hospital full?**

A) 50%

B) 53%

C) 52%

✓ D) 80%



**Q118. Department used 765g chlorine lime vs. avg 500g. How many % more bleach?**

A) 50%

✓ B) 53%

C) 52%

D) 60%

**Q119. Nurse had 25 packs sterile napkins. Used 1 pack. What % was used?**

A) 5%

✓ B) 4%

C) 52%

D) 60%

**Q120. Volume of cerebrospinal fluid: length h=43cm, diameter d=2cm. Volume?**

A) 132.01 cm<sup>3</sup>

B) 136.03 cm<sup>3</sup>

✓ C) 135.02 cm<sup>3</sup>

D) 134.05 cm<sup>3</sup>

**Q121. Transporting drugs: 2% ampoules broken = 5 pieces. Total ampoules?**

A) 620

B) 340

C) 560

✓ D) 250

**Q122. Brain of adult=1370g, newborn=400g. What % of adult brain is newborn brain?**

✓ A) 29.2%

B) 35.5%

C) 70.7%

D) 10.3%

**Q123. Skeleton: 208 bones, 85 paired. How many unpaired?**

A) 95

✓ B) 38

C) 83

D) 73

**Q124. Blood volume adult=5.5L. Deep cut loses 8%. Amount of blood loss?**

✓ A) 0.44 L

B) 0.3 L

C) 4 L

D) 1 L

**Q125. Microsoft Access is a:**

A) Network Database Model

✓ B) RDBMS

C) ORDBMS

D) OODBMS

**Q126. Which is NOT a type of Microsoft Access database object?**

A) Macros

B) Modules

✓ C) Worksheets

D) Table

**Q127. A subset of characters within a data field is known as:**

A) Byte

B) File

C) Record

✓ D) Data string

**Q128. Press \_\_\_\_\_ to quit MS Access.**

- A) Tab + F4
- B) Esc + W
- C) Ctrl + F4

✓ D) Alt + F4

**Q129. \_\_\_\_\_ is NOT a valid data type in MS Access.**

- A) Auto number
- B) Currency
- C) Memo

✓ D) Picture

**Q130. A \_\_\_\_\_ is an area reserved for a specific piece of data.**

- A) Report
- B) Key
- C) Record

✓ D) Field

**Q131. What is the maximum length a text field can be?**

- A) 75
- B) 120

✓ C) 255

- D) 265

**Q132. In Access, \_\_\_\_\_ are used to store the data.**

- A) Report
- B) Form

✓ C) Table

- D) Query

**Q133. Which tool do you use to create a query object?**

- A) Table query wizard
- ✓ B) Simple query wizard
- C) Simple filter wizard
- D) Database wizard

**Q134. Open the Save As dialog box:**

- A) F1
- B) F2
- C) F10
- ✓ D) F12

**Q135. In one-to-many relationship, the 'one' side is called \_\_\_\_\_ and 'many' side is called \_\_\_\_\_.**

- A) Father, Son
- B) Brother, Sister
- ✓ C) Parent, Child
- D) Child, Parent

**Q136. To create a table in MS Access:**

- A) Database should be created before creating a table
- B) You should allow at least one Primary key in a table
- C) Table can be created using Table Templates
- ✓ D) All of the above

**Q137. Database access levels are specified through:**

- A) User ID
- B) Password
- C) Status
- ✓ D) All of these

**Q138. The default and maximum size of text field in Access:**

✓ A) 50 and 255 characters

B) 266 and 6400 characters

C) 288 and 6880 characters

D) 299 and 6499 characters

**Q139. If  $A = \{x|x \in \mathbb{N}, x \leq 6\}$  and  $B = \{x|x \in \mathbb{N}, x \leq 10\}$ , find  $A \cap B$ . [Q164]**

A)  $\{1,2,3,4,5,6\}$

✓ B)  $\{1,2,3,4,5\}$

C)  $\{1,2,3,\dots,10\}$

D)  $\{1,2,\dots,9\}$

**Q140. If  $A = \{x|x \in \mathbb{N}, 4 \leq x \leq 9\}$  and  $B = \{x|x \in \mathbb{N}, 4 \leq x \leq 10\}$ , find  $A \cap B$ . [Q165]**

A)  $\{4,5,6,7,8,9\}$

✓ B)  $\{5,6,7,8\}$

C)  $\{9\}$

D)  $\{4,\dots,10\}$

**Q141. If  $A = \{x|x = 2n, n \in \mathbb{N}, 4 \leq n \leq 12\}$  and  $B = \{x|x = 3n, n \in \mathbb{N}, 3 \leq n \leq 18\}$ , find  $A \cup B$ . [Q166]**

A) 3 to 18

B) 6 to 15

✓ C)  $\{6,8,9,10,12,15\}$

D) 4 to 18

**Q142. If  $A = \{x|x \in \mathbb{Z}, x \geq -4\}$  and  $B = \{x|x \in \mathbb{Z}, x \leq 3\}$ , find  $A \cap B$ . [Q167]**

✓ A)  $\{-3,-2,-1,0,1,2\}$

B)  $\{-4,\dots,3\}$

C)  $\{-4,3\}$

D) 0

**Q143. If  $A=\{x|x \in \mathbb{N}, x \geq 2\}$  and  $B=\{x|x \in \mathbb{Z}, x \leq 7\}$ , find  $A \cap B$ . [Q168]**

A) 2,3,4,5,6,7

B) 5

✓ C) {3,4,5,6}

D)  $-\infty$  to  $+\infty$

**Q144. If  $A=\{x|x \in \mathbb{N}, x \geq 2\}$  and  $B=\{x|x \in \mathbb{Z}, x \leq 7\}$ , find  $A \cup B$ . [Q169]**

A) 2,3,4,5,6,7

B) 5

C) {3,4,5,6}

✓ D)  $-\infty$  to  $+\infty$

**Q145. If  $A=\{x|x \in \mathbb{R}, -1 \leq x \leq 5\}$  and  $B=\{x|x \in \mathbb{R}, 1 \leq x \leq 6\}$ , find  $B/A$ . [Q170]**

A) {6}

✓ B) {5}

C) {0,1,...,6}

D) {-1,2,...,5}

**Q146. If  $A=\{x|x \in \mathbb{R}, -1 \leq x \leq 5\}$  and  $B=\{x|x \in \mathbb{R}, 1 \leq x \leq 6\}$ , find  $A \cup B$ . [Q172]**

✓ A) -1 to 6

B) 0 to 6

C) 0 to 5

D) {5,6}

**Q147. If  $A=\{x|x=2n, n \in \mathbb{N}, 4 \leq n \leq 12\}$  and  $B=\{x|x=3n, n \in \mathbb{N}, 3 \leq n \leq 18\}$ , find  $A \cap B$ . [Q173]**

✓ A) {6,12}

B) {4,12}

C) {3,12}

D)  $\emptyset$

**Q148. Find the derivative  $y = x^2 \cos(x)$ . [Q174]**

- A)  $2x \sin x$
- B)  $-2x \sin x$
- C)  $2x \cos x + x^2 \sin x$

✓ D)  $2x \cos x - x^2 \sin x$

**Q149. Find the derivative  $y = \sin(6x-1.5)$ . [Q175]**

- A)  $\cos(6x-1.5)$
- B)  $-9\cos(6x-15)$

✓ C)  $6\cos(6x-1.5)$

- D)  $-\cos(6x-1.5)$

**Q150. Find the derivative  $y = \sin(3x+2) + \cos 9 + 1000$ . [Q176]**

- A)  $\cos(3x+2) - \sin 9$
- B)  $-3\cos(3x+2)\sin 9 + 1000x$

✓ C)  $3\cos(3x+2)$

- D)  $-\cos(3x+2) + \sin 9$

**Q151. Find the derivative  $y = \cos(5x-2) - \ln 10$ . [Q177]**

- A)  $-2\sin(5x-2) - 1/x$

✓ B)  $-5\sin(5x-2)$

- C)  $5\sin(5x-2) + \ln 10$
- D)  $\sin(5x-2)$

**Q152. Find the antiderivative F for  $f(x) = x^2 - \sin 2x$ . [Q178]**

- A)  $x^3/3 + \cos 2x + C$

✓ B)  $x^3/3 + \cos 2x/2 + C$

- C)  $2x - \sin 2x/2 + C$
- D)  $2x - \cos 2x/2 + C$

**Q153. Find the derivative  $y = (\sin 4x)/4$ . [Q181]**

A)  $\cos 4x/4$

✓ B)  $\cos 4x$

C)  $4\cos 4x$

D)  $\sin x$

**Q154. Find the derivative  $y = (x - 5x^2)e^x$ . [Q182]**

A)  $(1-10x)e^x$

B)  $(1-10x)e^x - (x-5x^2)e^x$

✓ C)  $(1-10x)e^x + (x-5x^2)e^x$

D)  $(1-5x)e^x$

**Q155. What is called integration? [Q183]**

✓ A) The operation of finding the integral

B) Transformation of an expression with integrals

C) The operation of finding the derivative

D) The limit of function increment to increment of argument

**Q156. What is the integration segment? [Q184]**

A) A circular area where the integral exists

✓ B) The interval over which the function needs to be integrated

C) The roots of the existence of the integrand

D) Integrand function

**Q157. Before Newton-Leibniz formula, which method was used? [Q190]**

A) Method of reduction to tabular integrals

✓ B) Transition to the limit of integral sums

C) Method of geometric transformations

D) Dirichlet method



**Q158. With which formula are definite integrals solved? [Q191]**

- A) Riemann's formulas
- B) Cauchy formulas
- C) Integral transformation formulas

✓ D) Newton-Leibniz formula

**Q159. Find the derivative  $y = \sin(x)/x + \cos(x)$ . [Q192]**

✓ A)  $(\cos(x) \cdot x - \sin(x))/x^2 - \sin(x)$

- B)  $\cos(x)/x + \sin(x)/x^2 - \sin(x)$
- C)  $\cos(x)/1 - \sin(x)$
- D)  $\cos(x) - \sin(x)$

**Q160. Solve the integral:  $\int 24dx/(x-3)$ . [Q193]**

A)  $24\ln x + 24\ln 3 + C$

✓ B)  $24\ln(x-3) + C$

- C)  $1/3 + C$
- D)  $24\ln/(x-3) + C$

**Q161. Solve the integral:  $\int (5+3x)^7 dx$ . [Q194]**

- A)  $7(5+3x)^6$
- B)  $(5+3x)^8/8 + C$
- C)  $(1/21)(5+3x)^8 + C$

✓ D)  $(1/24)(5+3x)^8 + C$

**Q162. Solve the integral:  $\int (\cos 12x + \ln 5) dx$ . [Q195]**

- A)  $\sin 12x + \ln 5x + C$
- B)  $-\sin 12x + \ln 5 + C$

✓ C)  $(1/12)\sin 12x + \ln 5x + C$

D)  $\sin 12x + C$

**Q163. Find the derived functions  $f(x) = 4\log_5(x) + \log_5(6) - 3/x$ . [Q196]**

✓ A)  $4/(x \ln 5) + 3/x^2$

B)  $1/(x \ln 5) + 3/x^2$

C)  $4/(x \ln 5) + 1/(6 \ln 5) - 3/x^2$

D)  $4/(x \ln 5) - 3/x^2$

**Q164. Find the derived functions  $f(x) = 3x^2 \ln(x) + e/2$ . [Q197]**

✓ A)  $3x(2\ln x + 1)$

B)  $6x \ln x + 3x^2 + 1/2$

C)  $6x \ln x + 3x$

D)  $2\ln x + 1$

**Q165. Find the indefinite integral  $\int 9e^t dt$ . [Q200]**

✓ A)  $9e^t + C$

B)  $e^t + C$

C)  $9 + C$

D)  $9e^t + 9 + C$

**Q166. Patient weighs 60 kg. Prescribed dose: 5 mg/kg. Total dose?**

A) 200 mg

✓ B) 300 mg

C) 250 mg

D) 350 mg

**Q167. Patient needs 3 doses of 50 mg each. Total medication?**

A) 100 mg

✓ B) 150 mg

C) 200 mg

D) 120 mg

**Q168. Solution contains 200 mg in 100 ml. How many mg in 250 ml?**

- A) 400 mg
- B) 450 mg
- ✓ C) 500 mg
- D) 300 mg

**Q169. Medicine requires 2 mg/kg for a 45 kg patient. Dose?**

- A) 80 mg
- ✓ B) 90 mg
- C) 100 mg
- D) 95 mg

**Q170. Pediatric patient weighs 15 kg. Prescribed 0.5 mg/kg. Total dose?**

- ✓ A) 7.5 mg
- B) 10 mg
- C) 6 mg
- D) 5 mg

**Q171. Patient prescribed 150 mg/day; tablet contains 50 mg each. Tablets/day?**

- A) 2
- ✓ B) 3
- C) 4
- D) 5

**Q172. IV drip: 500 ml over 5 hours. Flow rate (ml/hour)?**

- A) 90 ml/h
- ✓ B) 100 ml/h
- C) 80 ml/h
- D) 110 ml/h

**Q173. Patient needs 0.25 mg/kg. Weight: 80 kg. Total dose?**

A) 20 mg

✓ B) 25 mg

C) 30 mg

D) 22 mg

**Q174. Dose is 10 mg/kg. Patient weighs 70 kg. Total dose?**

✓ A) 700 mg

B) 600 mg

C) 650 mg

D) 750 mg

**Q175. Medicine prescribed 0.8 mg/kg. Patient weight: 50 kg. Total dose?**

A) 40 mg

B) 50 mg

✓ C) 40 mg (correct)

D) 45 mg

**Q176. Patient needs 200 mg/day in 4 doses. Amount per dose?**

A) 40 mg

✓ B) 50 mg

C) 60 mg

D) 45 mg

**Q177. Tablet contains 25 mg. Patient dose: 75 mg. Number of tablets?**

A) 2

✓ B) 3

C) 4

D) 5

**Q178. Prescribed 1.5 mg/kg. Patient weighs 20 kg. Dose?**

A) 25 mg

✓ B) 30 mg

C) 35 mg

D) 28 mg

**Q179. Medicine: 0.2 mg/kg. Weight: 65 kg. Dose?**

A) 12 mg

B) 13 mg

✓ C) 13 mg

D) 14 mg

**Q180. Patient prescribed 400 mg/day. Tablet: 100 mg. How many tablets/day?**

A) 3

✓ B) 4

C) 5

D) 6

**Q181. IV infusion: 1000 ml over 8 hours. Flow rate?**

✓ A) 120 ml/h

B) 110 ml/h

C) 100 ml/h

D) 105 ml/h

**Q182. Dose: 2 mg/kg. Weight: 35 kg. Total dose?**

A) 60 mg

✓ B) 70 mg

C) 65 mg

D) 75 mg

**Q183. Tablet: 50 mg. Prescribed 125 mg. Number of tablets?**

A) 2

✓ B) 2.5

C) 3

D) 3.5

**Q184. Patient needs 0.4 mg/kg. Weight: 45 kg. Dose?**

✓ A) 18 mg

B) 20 mg

C) 15 mg

D) 16 mg

**Q185. IV drip: 750 ml over 6 hours. Flow rate?**

A) 100 ml/h

✓ B) 125 ml/h

C) 120 ml/h

D) 130 ml/h

**Q186. Dose: 5 mg/kg. Weight: 30 kg. Total dose?**

✓ A) 150 mg

B) 140 mg

C) 130 mg

D) 160 mg

**Q187. Tablet contains 20 mg. Prescribed 60 mg. Number of tablets?**

A) 2

✓ B) 3

C) 4

D) 5

**Q188. Pediatric dose: 0.6 mg/kg. Weight: 25 kg. Dose?**

A) 12 mg

✓ B) 15 mg

C) 14 mg

D) 16 mg

**Q189. Patient prescribed 300 mg/day in 3 doses. Amount per dose?**

A) 90 mg

✓ B) 100 mg

C) 110 mg

D) 120 mg

**Q190. IV infusion: 600 ml over 4 hours. Flow rate?**

A) 140 ml/h

✓ B) 150 ml/h

C) 160 ml/h

D) 145 ml/h

**Q191. Solution contains 250 mg drug in 500 ml. Concentration?**

A) 0.25%

✓ B) 0.5%

C) 1%

D) 2%

**Q192. Mix 1:4 ratio of medicine to water. How much medicine in 100 ml solution?**

✓ A) 20 ml

B) 25 ml

C) 15 ml

D) 10 ml

**Q193. 200 mg drug dissolved in 100 ml solution. Percentage concentration?**

A) 0.2%

B) 0.5%

✓ C) 0.2% (correct)

D) 2%

**Q194. IV solution: 5 mg in 50 ml. Concentration?**

✓ A) 0.5%

B) 1%

C) 0.1%

D) 0.05%

**Q195. Solution contains 300 mg in 600 ml. Concentration?**

A) 0.05%

✓ B) 0.1%

C) 0.5%

D) 1%

**Q196. Mix 2:3 drug to water. Total volume 50 ml. Amount of drug?**

✓ A) 20 ml

B) 25 ml

C) 30 ml

D) 15 ml

**Q197. Patient needs 0.2 g medicine in 250 ml. Percentage?**

A) 0.08%

✓ B) 0.08% (correct)

C) 0.2%

D) 0.02%



**Q198. Solution: 400 mg in 800 ml. Percent concentration?**

A) 0.05%

✓ B) 0.1%

C) 0.5%

D) 1%

**Q199. Drug concentration: 5 mg/50 ml. How many mg in 200 ml?**

A) 15 mg

✓ B) 20 mg

C) 25 mg

D) 30 mg

**Q200. 1:9 solution of drug to water. How much drug in 90 ml solution?**

✓ A) 10 ml

B) 9 ml

C) 15 ml

D) 12 ml

**Q201. Solution: 600 mg in 300 ml. Percentage concentration?**

A) 0.2%

B) 0.5%

C) 1%

✓ D) 2%

**Q202. Mix 3:7 drug to water. Total volume 100 ml. Drug amount?**

✓ A) 30 ml

B) 20 ml

C) 25 ml

D) 35 ml

**Q203. Patient requires 0.1 g in 200 ml. Percent concentration?**

✓ A) 0.05%

B) 0.1%

C) 0.2%

D) 0.02%

**Q204. Solution contains 50 mg in 100 ml. Percent concentration?**

✓ A) 0.05%

B) 0.5%

C) 1%

D) 0.005%

**Q205. Mix ratio 1:3, total solution 80 ml. Amount of drug?**

✓ A) 20 ml

B) 25 ml

C) 15 ml

D) 30 ml

**Q206. Solution: 0.5 g in 250 ml. Percent concentration?**

A) 0.1%

✓ B) 0.2%

C) 0.5%

D) 1%

**Q207. IV solution: 200 mg in 400 ml. Percent concentration?**

A) 0.05%

✓ B) 0.1%

C) 0.2%

D) 0.5%

**Q208. Mix 2:8 drug to water. Total 100 ml. Drug amount?**

A) 15 ml

✓ B) 20 ml

C) 25 ml

D) 10 ml

**Q209. Patient needs 0.3 g in 150 ml. Percent concentration?**

A) 0.1%

✓ B) 0.2%

C) 0.15%

D) 0.3%

**Q210. Solution: 250 mg in 250 ml. Percent concentration?**

✓ A) 0.1%

B) 0.2%

C) 0.5%

D) 1%

**Q211. Patient temperatures: 36.8, 37.0, 37.2, 37.4. Mean temperature?**

✓ A) 37.1

B) 37.0

C) 36.9

D) 37.2

**Q212. Blood pressure readings: 120, 130, 125, 140, 135. Median?**

✓ A) 130

B) 125

C) 135

D) 140

**Q213. Heart rates: 72, 75, 70, 78, 74. Mode?**

A) 72

B) 75

✓ C) No mode

D) 74

**Q214. Temperatures: 36.5, 37.0, 37.0, 36.8, 37.2. Mode?**

A) 36.8

✓ B) 37.0

C) 36.5

D) 37.2

**Q215. Patient weights (kg): 60, 65, 70, 75, 80. Range?**

A) 15

✓ B) 20

C) 10

D) 25

**Q216. Blood sugar: 90, 95, 100, 85, 110. Mean?**

A) 95

✓ B) 96

C) 100

D) 98

**Q217. Heart rate readings: 70, 72, 68, 74, 70. Mode?**

✓ A) 70

B) 72

C) 68

D) 74

**Q218. Temperatures: 36.6, 36.8, 36.6, 37.0, 36.9. Mean?**

✓ A) 36.78

B) 36.8

C) 36.7

D) 36.9

**Q219. Blood pressure: 120, 125, 130, 135, 140. Median?**

A) 125

✓ B) 130

C) 135

D) 120

**Q220. Cholesterol readings (mg/dL): 180, 200, 220, 210, 190. Mean?**

✓ A) 200

B) 205

C) 210

D) 195

**Q221. Patient heights (cm): 160, 165, 170, 175, 180. Range?**

✓ A) 20

B) 15

C) 25

D) 30

**Q222. Heart rates: 68, 72, 70, 72, 74. Mode?**

A) 70

✓ B) 72

C) 68

D) 74

**Q223. Temperatures: 36.7, 36.9, 37.0, 36.8, 36.9. Mean?**

✓ A) 36.86

B) 36.9

C) 36.8

D) 36.85

**Q224. Blood sugar: 85, 90, 95, 100, 105. Median?**

A) 90

✓ B) 95

C) 100

D) 85

**Q225. Patient weights (kg): 55, 60, 65, 60, 70. Mode?**

A) 55

✓ B) 60

C) 65

D) 70

**Q226. Heart rates: 68, 70, 72, 74, 76. Range?**

A) 8

✓ B) 10

C) 12

D) 6

**Q227. Temperatures: 36.8, 36.7, 36.9, 36.8, 37.0. Mode?**

A) 36.7

✓ B) 36.8

C) 36.9

D) 37.0

**Q228. Blood pressure: 120, 125, 130, 120, 135. Mode?**

✓ A) 120

B) 125

C) 130

D) 135

**Q229. Patient weights: 50, 55, 60, 65, 70. Mean?**

✓ A) 60

B) 55

C) 65

D) 62

**Q230. Temperatures: 36.6, 36.8, 37.0, 36.9, 36.7. Range?**

✓ A) 0.4

B) 0.5

C) 0.3

D) 0.6

**Q231. Heart rates: 70, 72, 74, 72, 76. Mode?**

✓ A) 72

B) 70

C) 74

D) 76

**Q232. Blood sugar: 90, 95, 100, 95, 105. Mode?**

A) 90

✓ B) 95

C) 100

D) 105

**Q233. Temperatures: 36.8, 36.7, 36.9, 36.8, 36.7. Mean?**

✓ A) 36.78

B) 36.8

C) 36.75

D) 36.7

**Q234. Blood pressure: 110, 115, 120, 125, 130. Median?**

A) 115

✓ B) 120

C) 125

D) 110

**Q235. Patient heights (cm): 160, 165, 170, 165, 175. Mode?**

A) 160

✓ B) 165

C) 170

D) 175

**Q236. Heart rate readings: 70, 72, 75, 73, 74. Max increase between which days?**

A) Day 1-2

✓ B) Day 2-3

C) Day 3-4

D) Day 4-5

**Q237. Patient glucose levels trend upward over 5 days. What does graph indicate?**

A) Decreasing glucose

✓ B) Increasing glucose

C) Stable glucose

D) Unclear trend



**Q238. Chart shows systolic BP: 120, 125, 130, 135, 140. Median?**

A) 125

✓ B) 130

C) 135

D) 120

**Q239. Blood sugar chart: 90, 95, 100, 105, 110 mg/dL. Range?**

A) 15

✓ B) 20

C) 25

D) 10

**Q240. Patient temp line graph: 36.8, 37.0, 37.2, 37.4, 37.3. Largest increase?**

✓ A) Day 1-2

B) Day 2-3

C) Day 3-4

D) Day 4-5

**Q241. Graph heart rates: 70, 72, 71, 73, 74. Average rate?**

✓ A) 72

B) 71

C) 73

D) 74

**Q242. Patient glucose: 80, 85, 90, 95, 100 mg/dL. Trend?**

✓ A) Increasing

B) Decreasing

C) Stable

D) Fluctuating

**Q243. Line chart BP: 120, 122, 125, 127, 130 mmHg. Smallest increase?**

✓ A) 120-122

B) 122-125

C) 125-127

D) 127-130

**Q244. Vitals temperatures: 36.7, 36.8, 36.9, 37.0, 36.9. Median?**

A) 36.8

✓ B) 36.9

C) 37.0

D) 36.7

**Q245. Heart rate trend: 68, 70, 72, 70, 74. Mode?**

A) 68

✓ B) 70

C) 72

D) 74

**Q246. Glucose levels: 85, 90, 95, 100, 105. Mean?**

✓ A) 95

B) 96

C) 97

D) 94

**Q247. Line graph patient weight: 60, 62, 64, 66, 68 kg. Range?**

A) 6

✓ B) 8

C) 10

D) 12

**Q248. Chart heart rates: 70, 72, 74, 72, 76. Median?**

✓ A) 72

B) 74

C) 70

D) 76

**Q249. Blood sugar over a week: 90, 92, 95, 97, 100. Max increase?**

A) Day 1-2

✓ B) Day 2-3

C) Day 3-4

D) Day 4-5

**Q250. Temperatures graph: 36.6, 36.7, 36.8, 36.7, 36.9. Mode?**

A) 36.6

✓ B) 36.7

C) 36.8

D) 36.9

**Q251. Patient receives 3 doses: 150mg, 200mg, 250mg. Average dose?**

✓ A) 200 mg

B) 180 mg

C) 250 mg

D) 150 mg

**Q252. IV drip: 500 ml to run over 5 hours. Flow rate (ml/hour)?**

A) 50 ml/h

✓ B) 100 ml/h

C) 120 ml/h

D) 150 ml/h

**Q253. 250 mg medication divided into 5 equal doses. How much per dose?**

A) 40 mg

✓ B) 50 mg

C) 60 mg

D) 45 mg

**Q254. Patient weighs 70 kg. Prescribed 1 mg/kg. Total dose?**

✓ A) 70 mg

B) 60 mg

C) 65 mg

D) 75 mg

**Q255. IV infusion: 1000 ml over 10 hours. Flow rate?**

A) 80 ml/h

✓ B) 100 ml/h

C) 120 ml/h

D) 90 ml/h

**Q256. Patient needs 0.5 g medicine daily in 2 doses. Amount per dose?**

✓ A) 0.25 g

B) 0.2 g

C) 0.3 g

D) 0.4 g

**Q257. Tablet contains 50 mg. Patient dose: 150 mg. How many tablets?**

A) 2

✓ B) 3

C) 4

D) 5

**Q258. Patient receives 5 doses of 60 mg. Total medication?**

A) 250 mg

✓ B) 300 mg

C) 280 mg

D) 350 mg

**Q259. IV solution: 500 ml contains 200 mg drug. How much drug in 125 ml?**

✓ A) 50 mg

B) 60 mg

C) 40 mg

D) 45 mg

**Q260. Patient weight: 45 kg. Dose: 2 mg/kg. Total dose?**

A) 80 mg

✓ B) 90 mg

C) 100 mg

D) 95 mg

**Q261. 0.3 g in 150 ml solution. Percent concentration?**

A) 0.1%

✓ B) 0.2%

C) 0.15%

D) 0.3%

**Q262. Patient requires 1 mg/kg/day. Weight: 60 kg. Dose per day?**

A) 50 mg

✓ B) 60 mg

C) 70 mg

D) 55 mg

**Q263. IV drip: 750 ml over 6 hours. Flow rate?**

- A) 120 ml/h
- ✓ B) 125 ml/h
- C) 130 ml/h
- D) 150 ml/h

**Q264. Patient receives 4 doses of 75 mg. Total dose?**

- ✓ A) 300 mg
- B) 280 mg
- C) 350 mg
- D) 250 mg

**Q265. Tablet: 25 mg. Patient dose: 100 mg. How many tablets?**

- A) 3
- ✓ B) 4
- C) 5
- D) 6

**Q266. An element of a set is denoted by the symbol:**

- A)  $\subset$
- ✓ B)  $\in$
- C)  $\cup$
- D)  $\emptyset$

**Q267. The empty set is denoted by:**

- A)  $\{0\}$
- ✓ B)  $\emptyset$
- C)  $\{\}$
- D)  $\{\emptyset\}$

**Q268. If  $A \subset B$ , then:**

- A) A is larger than B
- B) A is not related to B
- ✓ C) Every element of A belongs to B
- D) B belongs to A

**Q269. The union of sets is denoted by:**

- A)  $\cap$
- B)  $\subset$
- ✓ C)  $\cup$
- D)  $\in$

**Q270. The intersection of sets is denoted by:**

- A)  $\cup$
- ✓ B)  $\cap$
- C)  $\subset$
- D)  $\emptyset$

**Q271. If  $A = \{1,2,3\}$  and  $B = \{3,4\}$ , then  $A \cap B$  is:**

- A)  $\{1,2\}$
- ✓ B)  $\{3\}$
- C)  $\{4\}$
- D)  $\{1,2,3,4\}$

**Q272. If  $A = \{1,2\}$  and  $B = \{3,4\}$ , then  $A \cap B$  is:**

- ✓ A)  $\emptyset$
- B)  $\{1\}$
- C)  $\{2\}$
- D)  $\{3\}$

**Q273. The number of elements in a set is called:**

- A) element
- B) intersection
- ✓ C) cardinality
- D) subset

**Q274. The cardinality of the set  $A = \{a,b,c\}$  is:**

- A) 1
- B) 2
- ✓ C) 3
- D) 4

**Q275. The universal set is denoted by:**

- A) A
- B)  $\emptyset$
- ✓ C) U
- D)  $\cap$

**Q276. The difference of sets  $A \setminus B$  is:**

- A) Elements common to A and B
- B) Elements of B not in A
- ✓ C) Elements of A not in B
- D) The empty set

**Q277. If  $A \subset B$  and  $A \neq B$ , then A is:**

- A) An equal set
- ✓ B) A proper subset
- C) An empty set
- D) A universal set



**Q278. Is the statement  $\emptyset \subset A$  always true?**

- A) No
- B) Sometimes
- ✓ C) Yes
- D) Only if A is empty

**Q279. If  $A = B$ , then:**

- A)  $A \subset B$
- B)  $B \subset A$
- C) A and B have the same elements
- ✓ D) All answers are correct

**Q280. The symbol  $\not\subset$  means:**

- A) Belongs to
- B) Does not belong to
- ✓ C) Is not a subset
- D) Intersection

**Q281. If  $x \notin A$ , then:**

- A) x is an element of A
- ✓ B) x does not belong to A
- C) A is empty
- D) A is universal

**Q282. The set of natural numbers is denoted by:**

- A) Z
- B) Q
- ✓ C) N
- D) R

**Q283. The set of integers is denoted by:**

- A) N
- B) Q
- C) R

✓ D) Z

**Q284. Diagrams used to represent sets visually are called:**

- A) Graphs
- B) Tables

✓ C) Venn diagrams

- D) Coordinate diagrams

**Q285. If  $A \cap B = \emptyset$ , the sets are called:**

- A) Equal
- B) Intersecting

✓ C) Disjoint

- D) Universal

**Q286. If  $A = \{x \mid x > 0\}$ , then A is the set of:**

- A) Negative numbers

✓ B) Positive numbers

- C) Integers
- D) Rational numbers

**Q287. A subset can be:**

- A) Only smaller than the set
- B) Only larger than the set

✓ C) Equal to the set

- D) Unrelated to the set

**Q288.  $A \cup A$  equals:**A)  $\emptyset$ **✓ B)  $A$** C)  $U$ D)  $B$ **Q289.  $A \cap A$  equals:**A)  $\emptyset$ B)  $U$ **✓ C)  $A$** D)  $B$ **Q290.  $A \cup \emptyset$  equals:**A)  $\emptyset$ **✓ B)  $A$** C)  $U$ D)  $\{0\}$ **Q291.  $A \cap \emptyset$  equals:**A)  $A$ B)  $U$ **✓ C)  $\emptyset$** D)  $\{A\}$ **Q292. If  $A \subset U$ , then  $U$  is:**A)  $A$  subset

B) An empty set

**✓ C) A universal set**

D) An intersection

**Q293. A finite set is a set that:**

- A) Has no elements
- B) Has infinitely many elements
- ✓ C) Has a limited number of elements
- D) Is universal

**Q294. An infinite set is a set that:**

- A) Is empty
- B) Has a finite number of elements
- ✓ C) Has no last element
- D) Equals the universal set

**Q295.  $\log_a(a)$  equals:**

- A) 0
- ✓ B) 1
- C) a
- D) undefined

**Q296. The base of a logarithm must be:**

- A) Negative
- B) Zero
- ✓ C) Positive and not equal to 1
- D) Any real number

**Q297.  $\log_{10}(10) =$**

- A) 0
- ✓ B) 1
- C) 10
- D) 2

**Q298.  $\log_a(xy)$  equals:**

✓ A)  $\log_a(x) + \log_a(y)$

B)  $\log_a(x) - \log_a(y)$

C)  $\log_a(x-y)$

D)  $\log_a(x/y)$

**Q299.  $\log_a(x/y)$  equals:**

A)  $\log_a(x) + \log_a(y)$

✓ B)  $\log_a(x) - \log_a(y)$

C)  $\log_a(xy)$

D)  $\log_a(x-y)$

**Q300.  $\log_a(x^n)$  equals:**

✓ A)  $n \log_a(x)$

B)  $\log_a(x^n)$

C)  $\log_a(x) + n$

D)  $\log_a(x)/n$

**Q301.  $\log_2(8)$  equals:**

A) 2

✓ B) 3

C) 4

D) 8

**Q302. The logarithmic form of  $2^3 = 8$  is:**

✓ A)  $\log_2(8) = 3$

B)  $\log_8(2) = 3$

C)  $\log_3(2) = 8$

D)  $\log_2(3) = 8$

**Q303. The natural logarithm is denoted by:**

A)  $\log$

✓ B)  $\ln$

C)  $\lg$

D)  $\exp$

**Q304. The base of the natural logarithm is:**

A) 2

B) 10

✓ C)  $e$

D) 1

**Q305.  $\ln(e)$  equals:**

A) 0

✓ B) 1

C)  $e$

D) undefined

**Q306.  $\ln(1)$  equals:**

A) 1

B)  $e$

✓ C) 0

D) undefined

**Q307. Which expression is undefined?**

A)  $\log_2(8)$

B)  $\log_3(1)$

✓ C)  $\log_{10}(0)$

D)  $\ln(e)$

**Q308. The domain of  $y = \log_a(x)$  is:**

A)  $x \geq 0$

✓ B)  $x > 0$

C) All real numbers

D)  $x \leq 0$

**Q309. The graph of  $y = \log_a(x)$  passes through the point:**

A) (0,1)

✓ B) (1,0)

C) (1,1)

D) (0,0)

**Q310. If  $a > 1$ , the logarithmic function is:**

A) Decreasing

B) Constant

✓ C) Increasing

D) Undefined

**Q311. If  $0 < a < 1$ , the logarithmic function is:**

A) Increasing

✓ B) Decreasing

C) Constant

D) Linear

**Q312.  $\log_{10}(1000)$  equals:**

A) 2

✓ B) 3

C) 10

D) 100

**Q313.  $\log_5(25)$  equals:**

✓ A) 2

B) 5

C) 25

D) 1

**Q314. Which is true?**

A)  $\log_a(x+y) = \log_a(x) + \log_a(y)$

✓ B)  $\log_a(xy) = \log_a(x) + \log_a(y)$

C)  $\log_a(x-y) = \log_a(x) - \log_a(y)$

D)  $\log_a(x+y) = \log_a(x) - \log_a(y)$

**Q315. The inverse function of  $y = \log_a(x)$  is:**

A)  $y = ax$

✓ B)  $y = a^x$

C)  $y = x^a$

D)  $y = x/a$

**Q316.  $\log_2(1)$  equals:**

A) 1

B) 2

✓ C) 0

D) undefined

**Q317.  $\log_4(16)$  equals:**

✓ A) 2

B) 4

C) 8

D) 16



**Q318. Which number is NOT allowed as a logarithm base?**

- A) 2
- B) 10
- C) e

✓ D) 1

**Q319. The expression  $\log_3(-9)$  is:**

- A) 2
- B) -2
- C) 0

✓ D) Undefined

**Q320. If  $\log_a(x) = 0$ , then x equals:**

- A) 0

✓ B) 1

- C) a
- D) -1

**Q321. Logarithms are used to:**

- A) Simplify multiplication and division
- B) Solve exponential equations
- C) Analyze exponential growth

✓ D) All answers are correct

**Q322. The derivative of  $f(x) = x^2 - 4x$  is:**

✓ A)  $2x - 4$

- B)  $x - 4$
- C)  $2x$
- D)  $4x$

**Q323. The derivative of  $f(x) = \ln(3x)$  is:**

A)  $3/x$

✓ B)  $1/x$

C)  $\ln x$

D)  $3$

**Q324. The derivative of  $f(x) = \tan x$  is:**

A)  $\cos x$

B)  $\sin x$

✓ C)  $\sec^2(x)$

D)  $-\sec^2(x)$

**Q325. The derivative of  $f(x) = x^2 \sin x$  is:**

A)  $2x \sin x$

B)  $x^2 \cos x$

✓ C)  $2x \sin x + x^2 \cos x$

D)  $\sin x + \cos x$

**Q326. The derivative of  $f(x) = e^{(2x)}$  is:**

A)  $e^{(2x)}$

✓ B)  $2e^{(2x)}$

C)  $xe^{(2x)}$

D)  $2xe^{(2x)}$

**Q327. The derivative of  $f(x) = \ln(x^2)$  is:**

A)  $1/x$

✓ B)  $2/x$

C)  $x$

D)  $\ln x$

**Q328. The derivative of  $f(x) = 1/x^2$  is:**

- A)  $-1/x$
- ✓ B)  $-2/x^3$
- C)  $2/x^3$
- D)  $-1/x^2$

**Q329. The derivative of  $f(x) = \cos(3x)$  is:**

- A)  $-\sin(3x)$
- ✓ B)  $-3\sin(3x)$
- C)  $3\cos(3x)$
- D)  $\sin(3x)$

**Q330. The derivative of  $f(x) = x^3 e^x$  is:**

- A)  $3x^2 e^x$
- B)  $x^3 e^x$
- ✓ C)  $e^x(3x^2 + x^3)$
- D)  $x^2 e^x$

**Q331. The derivative of  $f(x) = \sqrt{x} \cdot \ln x$  is:**

- A)  $1/(2\sqrt{x})$
- B)  $\ln x/(2\sqrt{x})$
- ✓ C)  $(\ln x)/(2\sqrt{x}) + 1/\sqrt{x}$
- D)  $\sqrt{x}/x$

**Q332. The derivative of  $f(x) = e^x \sin x$  is:**

- A)  $e^x \cos x$
- B)  $e^x \sin x$
- ✓ C)  $e^x(\sin x + \cos x)$
- D)  $\sin x + \cos x$

**Q333. The derivative of  $f(x) = \ln(\sin x)$  is:**

- A)  $\cos x$
- B)  $\tan x$
- ✓ C)  $\cot x$
- D)  $1/\sin x$

**Q334. In medicine, derivatives help to analyze:**

- A) Rate of drug absorption
- B) Change of heart rate
- C) Tumor growth rate
- ✓ D) All answers are correct

**Q335. An integral represents:**

- A) Slope of a curve
- ✓ B) Area under a curve
- C) Maximum value
- D) Derivative

**Q336.  $\int x \, dx$  equals:**

- A)  $x^2 + C$
- ✓ B)  $x^2/2 + C$
- C)  $2x + C$
- D)  $x + C$

**Q337.  $\int 5 \, dx$  equals:**

- ✓ A)  $5x + C$
- B)  $x + C$
- C)  $5 + C$
- D)  $C$

**Q338.**  $\int x^2 dx$  equals:

A)  $x^3 + C$

✓ B)  $x^3/3 + C$

C)  $2x + C$

D)  $x^2/2 + C$

**Q339.** The constant of integration is denoted by:

A)  $x$

B)  $k$

✓ C)  $C$

D)  $a$

**Q340.**  $\int e^x dx$  equals:

✓ A)  $e^x + C$

B)  $xe^x + C$

C)  $\ln x + C$

D)  $1/e^x + C$

**Q341.**  $\int 1/x dx$  equals:

A)  $x + C$

✓ B)  $\ln|x| + C$

C)  $e^x + C$

D)  $1/x^2 + C$

**Q342.** An indefinite integral represents:

A) A number

✓ B) A function family

C) A limit

D) An equation

Q343.  $\int 3x^2 dx$  equals:

✓ A)  $x^3 + C$

B)  $3x^3 + C$

C)  $x^3/3 + C$

D)  $x^3 + C$

Q344.  $\int (2x + 1) dx$  equals:

✓ A)  $x^2 + x + C$

B)  $2x^2 + C$

C)  $x^2 + C$

D)  $x^2 + 1$

Q345.  $\int \sin x dx$  equals:

A)  $\cos x + C$

✓ B)  $-\cos x + C$

C)  $\sin x + C$

D)  $-\sin x + C$

Q346.  $\int \cos x dx$  equals:

✓ A)  $\sin x + C$

B)  $-\sin x + C$

C)  $\cos x + C$

D)  $-\cos x + C$

Q347.  $\int e^{(2x)} dx$  equals:

A)  $e^{(2x)} + C$

B)  $2e^{(2x)} + C$

✓ C)  $(1/2)e^{(2x)} + C$

D)  $e^x + C$

**Q348.**  $\int 1/\sqrt{x} \, dx$  equals:

A)  $\ln x + C$

✓ B)  $2\sqrt{x} + C$

C)  $\sqrt{x} + C$

D)  $-2\sqrt{x} + C$

**Q349.**  $\int (x^2 - 4x) \, dx$  equals:

A)  $x^3 - 2x^2 + C$

✓ B)  $x^3/3 - 2x^2 + C$

C)  $x^3 - 4x^2 + C$

D)  $x^2 - 2x + C$

**Q350.**  $\int \ln x \, dx$  equals:

✓ A)  $x \ln x - x + C$

B)  $\ln x + C$

C)  $x \ln x + C$

D)  $1/x + C$

**Q351.**  $\int x^{(-2)} \, dx$  equals:

✓ A)  $-x^{(-1)} + C$

B)  $x^{(-1)} + C$

C)  $-2x^{(-3)} + C$

D)  $x^{(-2)} + C$

**Q352.** A definite integral represents:

A) A family of functions

B) An equation

✓ C) A numerical value

D) A derivative

Q353.  $\int 1 x \, dx$  equals:

A) 1

✓ B)  $1/2$

C)  $1/3$

D) 2

Q354.  $\int 1.5 x \, dx$  equals:

A) 5

B) 6

✓ C) 7.5

D) 8

Q355. According to Newton-Leibniz formula:

✓ A)  $\int_a^b f(x) dx = F(b) - F(a)$

B)  $\int f(x) dx = F(x)$

C)  $d/dx \int f(x) dx = 0$

D)  $\int_a^a f(x) dx = 0$

Q356.  $\int (x e^x) \, dx$  equals:

A)  $e^x + C$

B)  $x e^x + C$

✓ C)  $e^x(x-1) + C$

D)  $x^2 e^x + C$

Q357.  $\int x \cos x \, dx$  equals:

A)  $\sin x + C$

✓ B)  $x \sin x - \cos x + C$

C)  $\cos x - x \sin x + C$

D)  $-x \sin x + C$



**Q358.**  $\int 1/(1+x^2) dx$  equals:

A)  $\ln x + C$

✓ B)  $\arctan x + C$

C)  $\sin x + C$

D)  $\cos x + C$

**Q359.**  $\int e^x \sin x dx$  equals:

✓ A)  $e^x(\sin x - \cos x)/2 + C$

B)  $e^x(\sin x + \cos x) + C$

C)  $e^x \cos x + C$

D)  $\sin x + C$

**Q360.**  $\int x^2 e^x dx$  equals:

A)  $x^2 e^x + C$

✓ B)  $e^x(x^2 - 2x + 2) + C$

C)  $e^x(x^2 + 2x + 2) + C$

D)  $e^x x^2 + C$

**Q361.** In medicine, integrals are used to calculate:

A) Total drug exposure over time

B) Accumulated radiation dose

C) Total blood flow

✓ D) All answers are correct